**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING**

**GOVERNMENT ENGINEERING COLLEGE, BARTON HILL**

**THIRUVANANTHAPURAM, KERALA**

# PROJECT REPORT ON

# TRASHBOT: An Intelligent Waste Collection and Segregation Robot

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# TRASHBOT: An Intelligent Waste Collection and Segregation Robot

*A Project Report*

*Submitted to the APJ Abdul Kalam Technological University*

*in partial fulfillment of requirements for the award of degree*

***Bachelor of Technology***

*in*

***Electronics and Communication Engineering***

*by*

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## CERTIFICATE

This is to certify that the report entitled **TrashBot: An Intelligent Waste Collection And Segregation Robot** is a bonafide record of work done by the team consisting of **Archita R** (TRV22EC014), **Hari Nanda B** (TRV22EC036), **Nived Rajeesh V P** (TRV22EC058) & **Yadhuna V S** (TRV22EC068).This report is submitted to the Department of **Electronics and Communication Engineering**, Govt. Engineering College, Barton Hill, in partial fulfillment for the award of B.Tech. degree in **Electronics and Communication Engineering** under APJ Abdul Kalam Technological University during the year 2025-26.

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**DECLARATION**

We hereby declare that the project report **TRASHBOT: An Intelligent Waste Collection and Segregation Robot** submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of Lekshmi T.

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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# Abstract

The escalating challenge of municipal solid waste management necessitates innovative, automated solutions to enhance efficiency and promote sustainability. Current methods of manual waste collection and segregation are often labor-intensive, unhygienic, and inefficient. To address this, we present Trashbot, an autonomous robot designed to intelligently collect and segregate waste in indoor environments.

The system leverages a Raspberry Pi as its central processing unit, integrating a LIDAR sensor for autonomous navigation and obstacle avoidance. A key feature of Trashbot is its machine learning-based waste classification system, which utilizes a Raspberry Pi camera and the YOLOv8 algorithm to visually identify and categorize waste into biodegradable and non-biodegradable types in real-time. Upon detection, a custom 3D-printed robotic arm, actuated by high-torque servo motors, executes the physical pickup and places the waste into the corresponding segregated bin.

This project successfully demonstrates the integration of robotics and artificial intelligence to create an end-to-end automated waste management system. Trashbot offers a practical solution to reduce human contact with waste, improve recycling rates through accurate segregation, and serves as a foundational model for smart and sustainable waste disposal systems in campuses, offices, and other public spaces.

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# List of Abbreviations

1. SOSMC - Second-Order Sliding Mode Controller
2. PID - Proportional Integral Derivative
3. PD - Proportional Derivative
4. MATLAB - Matrix Laboratory
5. MLRW - Magnetically Levitated Reaction Wheel
6. LOS - Line of Sight
7. AOCS - Attitude and Orbit Control System
8. PDF - Probability Density Function
9. AMB - Active Magnetic Bearing
10. EMC - Electromagnetic Compatibility
11. PWM - Pulse Width Modulation
12. SMC - Sliding Mode Controller
13. MPLAB - Microchip Programming Laboratory
14. LTSpice - Linear Technology Simulation Program with Integrated Circuit Emphasis

# Chapter 1

# Introduction

In recent years, waste management has become one of the most pressing challenges in both urban and indoor environments such as offices, hospitals, and educational institutions. The improper disposal of waste not only leads to environmental pollution but also poses significant health hazards to cleaning staff and the general public. Manual waste collection is often unhygienic, labor-intensive, and exposes workers to harmful contaminants, making automation in waste management an essential step toward cleaner and smarter spaces.

To address these challenges, TrashBot, a semi-autonomous robot, is proposed for efficient waste collection and segregation. The robot is designed to navigate indoor environments autonomously, detect waste using camera-based vision, and segregate it into biodegradable and non-biodegradable categories. By integrating robotics, computer vision, and machine learning, TrashBot aims to reduce human effort while ensuring hygiene and operational efficiency.

For navigation and mapping, the system employs LiDAR technology, which provides accurate distance measurement and obstacle detection, ensuring smooth and collision-free movement within confined indoor spaces. For waste classification, YOLOv8 (You Only Look Once version 8), an advanced real-time object detection algorithm, is utilized. YOLOv8 offers high-speed and high-accuracy detection, making it suitable for embedded systems and low-power computing platforms like ESP32 or Raspberry Pi.

The combination of LiDAR-based navigation and YOLOv8-based waste detection makes TrashBot a highly efficient, intelligent system capable of functioning in dynamic indoor environments. The robot's design also emphasizes energy efficiency, using a 12V lithium-ion battery with a buck converter to power various modules, ensuring compactness and portability.

Through this project, the goal is to demonstrate how embedded systems, sensors, and AI models can be integrated to create socially relevant solutions that support sustainability, cleanliness, and automation. TrashBot thus contributes to the development of intelligent waste management systems that align with smart infrastructure and environmental goals.

## 1.1 Summary

This project focuses on developing TrashBot, a smart indoor waste collection and segregation robot. The robot uses LiDAR for autonomous navigation and YOLOv8 for object detection and waste classification into biodegradable and non-biodegradable categories. Powered by an ESP32-based control system and a 12V Li-ion battery, the system ensures efficient, reliable, and hygienic waste management. By integrating robotics, computer vision, and embedded technology, the project aims to promote automation in waste handling, reduce human effort, and support sustainable and cleaner environments.

# Chapter 2

# Literature Review

H.-K. Chiang et al.[1] proposed a SOSMC method for robust stabilization and disturbance rejection in magnetic levitation systems, specifically for magnetic ball suspension. The authors demonstrate that SOSMC not only maintains the robustness typical of sliding mode control but also addresses chattering, a common drawback in traditional sliding mode control, by ensuring a continuous control signal. Experimental results validate that the SOSMC approach significantly enhances performance under uncertain conditions and external disturbances. This work forms a foundational basis for adopting SOSMC in our project to improve the stability and robustness of the levitation system.

The work by S. T. Spantideas and C. N. Capsalis [2] addresses the challenge of magnetic cleanliness in reaction wheels, focusing on the modeling of static and low-frequency magnetic fields. Through sensitivity analysis, they examine the key parameters influencing magnetic emissions and propose a methodology to model the magnetic signature of reaction wheels for applications requiring strict electromagnetic compatibility (EMC). This paper is pertinent to our prJ. de Heuvel et al. [1] proposed a spatiotemporal attention-based LiDAR navigation system for mobile robots operating in dynamic environments. The method enhances obstacle detection and real-time path planning accuracy by integrating temporal attention with LiDAR data. This improves robot adaptability in cluttered indoor settings, reducing collision risks. The findings from this study form the foundation for TrashBot's navigation mechanism, which uses LiDAR for smooth and autonomous indoor movement.

M. Nahiduzzaman et al. [2] developed an automated waste classification system using deep learning techniques to improve recycling efficiency and promote environmental sustainability. Their approach demonstrated high accuracy in classifying various waste types through convolutional neural networks (CNNs). The insights from this work guide the integration of computer vision and machine learning in TrashBot's waste segregation module.

M. Karthikeyan et al. [3] presented an SSD-based waste separation system employing augmented clustering NMS for enhanced waste detection. This system efficiently recognized and separated different waste categories in real time, offering inspiration for TrashBot's real-time visual waste recognition and decision-making.

R. Maurya et al. [4] introduced an AI-enabled system for classifying biodegradable and non-biodegradable materials using mobile applications. This research highlights the use of lightweight deep learning models for waste segregation, influencing TrashBot's design to use YOLOv8 for efficient and accurate classification of waste categories.

## 2.1 Summary

The reviewed literature explores advancements in robot navigation and waste classification using artificial intelligence and sensor fusion. LiDAR-based navigation ensures accurate path detection and obstacle avoidance, while machine learning models such as YOLOv8 and CNNs enable efficient waste identification. These studies collectively support the development of TrashBot as a smart, semi-autonomous indoor robot capable of precise navigation and real-time waste segregation into biodegradable and non-biodegradable categories, contributing to sustainable and hygienic waste management practices.

# Chapter 3

# Problem Statement

In indoor environments such as offices, hospitals, and educational institutions, improper waste disposal remains a major concern. Manual waste collection and segregation not only demand significant human effort but also expose cleaning staff to health hazards caused by direct contact with contaminated materials. Existing waste management methods often lack automation, resulting in inefficiency, inconsistency, and unhygienic handling of waste.

Current waste classification systems primarily depend on manual sorting or basic sensor-based techniques that cannot accurately distinguish between biodegradable and non-biodegradable waste. These approaches are prone to human error, limited scalability, and high maintenance, which restrict their practical implementation in smart environments. Similarly, robotic waste collectors that rely on traditional navigation systems using ultrasonic or infrared sensors face limitations in range, accuracy, and environmental adaptability.

To address these limitations, this project proposes the development of TrashBot, a semi-autonomous indoor robot equipped with LiDAR-based navigation and YOLOv8-based waste classification. The system combines advanced sensing, vision, and control technologies to enable real-time detection, collection, and segregation of waste. The use of YOLOv8 enhances classification accuracy, while LiDAR ensures efficient, collision-free navigation in cluttered indoor environments.

## 3.1 Existing Solutions

Several automated waste collection and classification systems have been developed using traditional sensors and simple computer vision models. Infrared (IR) and moisture sensors are commonly used for waste detection and segregation. While these solutions offer low-cost alternatives, they struggle with material ambiguity, lighting variations, and low classification accuracy.

Earlier robotic systems employed ultrasonic sensors for obstacle detection and navigation. Although effective in simple environments, they perform poorly in complex indoor areas where object shapes and materials vary. Machine learning-based waste classifiers using MobileNet or VGG architectures have achieved moderate accuracy but require significant computational resources, making them unsuitable for real-time embedded systems.

Recent research has shown promising results using YOLO-based architectures for waste classification. YOLOv8, in particular, offers superior speed and precision while maintaining computational efficiency, making it ideal for lightweight hardware such as ESP32 or Raspberry Pi. Similarly, LiDAR-based navigation systems have emerged as reliable solutions for indoor autonomous movement, outperforming traditional distance sensors in mapping accuracy and environmental awareness.

These advancements in object detection and navigation form the foundation for the TrashBot system, which integrates both approaches into a single, efficient platform for waste collection and segregation.

## 3.2 Objective

The objective of this project is to design and implement an autonomous indoor waste collection and segregation robot that can:

Navigate efficiently in indoor environments using LiDAR-based mapping and obstacle detection to ensure smooth and collision-free movement.

Identify and classify waste into biodegradable and non-biodegradable categories using the YOLOv8 algorithm, enabling accurate real-time segregation.

Integrate an efficient power system utilizing a 12V lithium-ion battery with a buck converter to provide stable power to all modules, ensuring extended operational time.

Develop a compact and efficient mechanical design with a single bin partitioned into two compartments for separate waste storage.

This system aims to reduce human intervention, enhance waste segregation accuracy, and promote hygiene and sustainability through intelligent automation.

## 3.3 Summary

This chapter presents the challenges associated with existing waste management systems, including manual handling, limited automation, and low classification accuracy. Traditional sensor-based methods often fail to address the complexity of waste materials and the dynamic nature of indoor environments. To overcome these issues, the proposed TrashBot integrates LiDAR navigation for precise movement and YOLOv8 for advanced waste classification. The system's objectives focus on developing a robust, efficient, and intelligent robotic platform capable of autonomously navigating and segregating waste into biodegradable and non-biodegradable categories, contributing to a smarter and cleaner environment.

# Chapter 4

# System Description

This chapter provides an in-depth explanation of the key components and interactions within the TrashBot system, focusing on its architecture and design methodology. Each subsystem — including navigation, detection, classification, and actuation — contributes to the robot's ability to autonomously collect and segregate waste. The integration of these subsystems ensures precise navigation, accurate waste identification, and reliable segregation. The chapter concludes with a summary of the overall system layout and the role of each component in achieving an efficient and intelligent waste management solution.

## 4.1 System Architecture

The system architecture of TrashBot comprises several integrated modules:

LiDAR-based navigation system,

camera and object detection unit,

waste collection mechanism, and

embedded control and power management system

The LiDAR sensor serves as the primary navigation component, continuously mapping the environment and detecting obstacles. Using the reflected laser pulses, it enables the robot to determine distances accurately, generate real-time spatial awareness, and plan collision-free paths within indoor spaces. This ensures smooth and efficient navigation in dynamic environments such as hospitals or campuses.

The camera module, interfaced with the ESP32 or Raspberry Pi, captures images of detected waste. These images are processed through the YOLOv8 object detection algorithm, which classifies the waste into biodegradable or non-biodegradable categories. The YOLOv8 model is optimized for lightweight deployment, making it suitable for real-time execution on low-power hardware platforms.

Once classification is complete, the actuator mechanism, powered by servo motors, directs the collected waste into the appropriate compartment within a partitioned single-bin system. This ensures on-the-spot segregation without requiring external sorting.

The power subsystem is built around a 12V lithium-ion battery, providing energy to all components. A buck converter regulates voltage levels to ensure safe and stable operation of microcontrollers and sensors. The control unit, based on the ESP32 microcontroller, coordinates data acquisition, processing, and actuation tasks in real time.

The entire system functions in a closed-loop configuration, where continuous feedback from the sensors ensures accurate decision-making and responsive behavior. The interaction between LiDAR, YOLOv8, and the control algorithms allows TrashBot to autonomously detect waste, avoid obstacles, and execute segregation operations seamlessly.

### 4.1.1 Robotic Arm

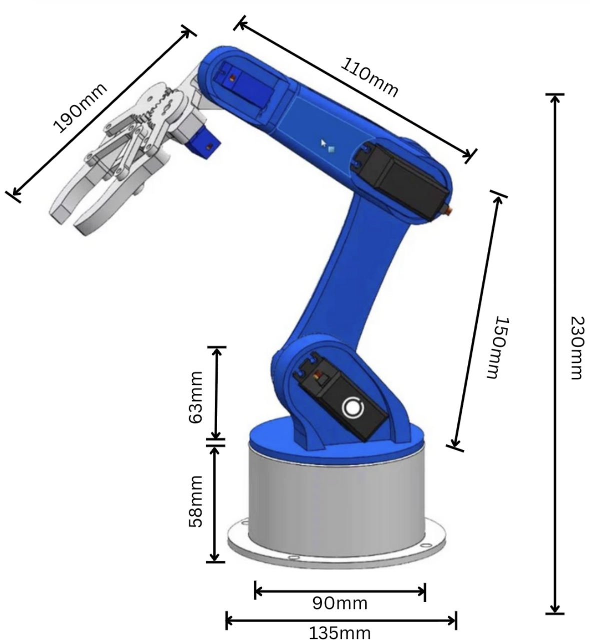


Figure 4.1: Robotic Arm

## 4.2 Summary

The TrashBot system architecture integrates LiDAR for navigation, YOLOv8 for waste classification, and a servo-based mechanism for waste segregation. The ESP32 microcontroller acts as the central control unit, processing sensor feedback and executing navigation and classification tasks. Power is supplied by a 12V lithium-ion battery with regulated output through a buck converter, ensuring stable operation. The system's closed-loop configuration enables real-time environmental perception, intelligent decision-making, and autonomous task execution. Together, these subsystems form a cohesive architecture capable of efficient, hygienic, and intelligent waste collection and segregation in indoor environments.

# Chapter 5

# System Design

This chapter discusses the design and implementation of the TrashBot system, focusing on both the hardware components and the software architecture. The hardware section outlines the key components required for autonomous navigation, waste detection, and segregation, while the software section describes the implementation of the YOLOv8 algorithm for intelligent waste classification.

## 5.1 Hardware Design

The hardware design of TrashBot includes a Raspberry Pi 5 as the central processing unit, a LiDAR sensor for navigation, a camera module for waste detection, DC motors for movement, servo motors for the robotic arm, and a motor driver for precise control. A buck converter regulates the power supply, and a custom PCB integrates all components. The system is powered by a pack of nine 12V lithium-ion batteries connected in series-parallel configuration to provide sufficient power for extended operation.

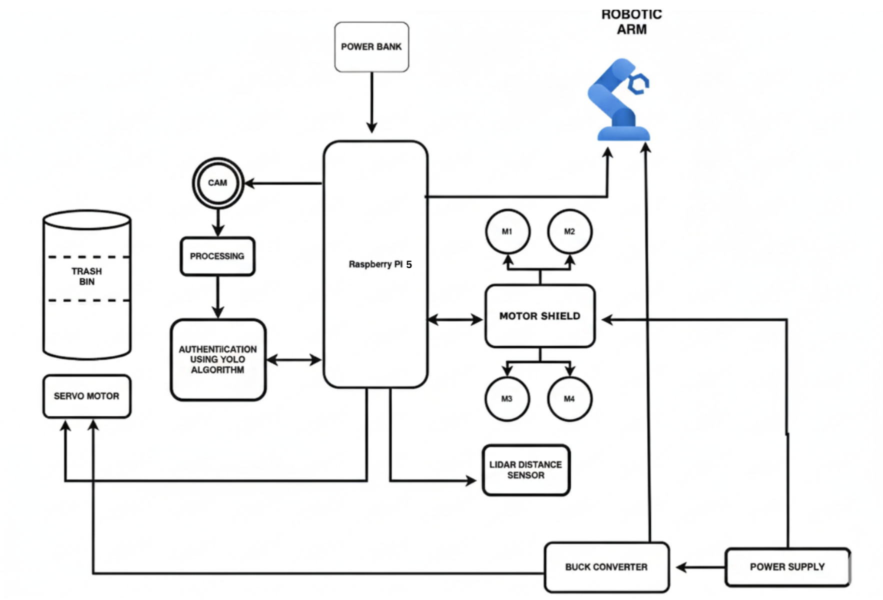


Figure 5.1: System block diagram of TrashBot

### 5.1.1 Raspberry Pi 5

The Raspberry Pi 5 serves as the main controller, processing data from the LiDAR and camera, and executing the YOLOv8 algorithm for waste detection and classification. With its enhanced processing capabilities and improved I/O performance, it efficiently handles real-time image processing and motor control simultaneously.

### 5.1.2 LiDAR Sensor

The system employs a VL53L0X Time-of-Flight (ToF) LiDAR sensor for autonomous navigation and obstacle detection. This sensor features:

- Measurement range: Up to 2 meters
- Accuracy: ±3% under optimal conditions
- Operating voltage: 2.6V to 3.5V
- Field of view: 25 degrees
- High precision distance measurement using Time-of-Flight technology

- Fast ranging capability up to 50Hz

The LiDAR enables the robot to create a real-time map of its surroundings, detect obstacles, and navigate autonomously within indoor environments such as corridors, offices, and institutional spaces.

### 5.1.3 Camera Module

The Raspberry Pi Camera Module v2 captures real-time images of waste objects with the following specifications:

- 8-megapixel Sony IMX219 image sensor
- Fixed focus lens
- CSI interface for high-speed data transfer

The captured images are processed by the YOLOv8 model to classify waste into biodegradable and non-biodegradable categories with high accuracy.

### 5.1.4 Motor Driver and DC Motors

The system utilizes four 12V DC gear motors with the following specifications:

- Speed: 60 RPM
- Torque: High torque for carrying payload
- Gear reduction for increased torque
- Encoder feedback for precise position control

These motors are controlled by an L298N motor driver that provides:

- Dual H-bridge configuration
- 2A maximum current per channel
- PWM speed control capability
- Thermal shutdown protection

The four-wheel configuration ensures stable movement and maneuverability in various indoor environments.

### 5.1.5 Servo Motors and Robotic Arm

The robotic arm system employs five MG996R high-torque servo motors with the following characteristics:

- Operating voltage: 4.8V to 7.2V
- Stall torque: 10 kg/cm at 6V
- Operating speed: 0.17s/60° at 6V
- Metal gear construction for durability

These servos provide the necessary torque and precision for the 3D-printed robotic arm to pick up waste items and place them in the appropriate bins.

### 5.1.6 Power Management

The power system consists of:

- Nine 12V lithium-ion batteries providing extended operational time
- LM2596 buck converter for voltage regulation
- Input voltage: 3V to 40V
- Output voltage: 1.5V to 35V (adjustable)
- Maximum output current: 3A
- Conversion efficiency: Up to 92%

This power management system ensures stable voltage supply to all components, including the Raspberry Pi 5 (5V), servo motors (6V), and DC motors (12V).

## 5.2 Software Design

The software design for TrashBot is centered around the Raspberry Pi 5, which runs the YOLOv8 object detection model trained on the COCO dataset. The system uses Python and OpenCV for image processing and real-time classification.

The YOLOv8 algorithm is employed for fast and accurate waste detection. It has been fine-tuned on a custom dataset derived from COCO to recognize and classify waste into biodegradable and non-biodegradable categories. The model processes images from the camera module and returns bounding boxes with class labels and confidence scores.

The control logic is implemented to coordinate between navigation (via LiDAR) and waste segregation (via the robotic arm). Once an object is detected and classified, the robotic arm is activated to pick the waste and place it in the corresponding bin.

## 5.3 Flowchart

The operational flowchart of TrashBot is illustrated in Figure 5.2. The process begins with object detection using LiDAR, followed by image capture and classification using YOLOv8. Based on the classification result, the robotic arm places the waste into the appropriate bin.

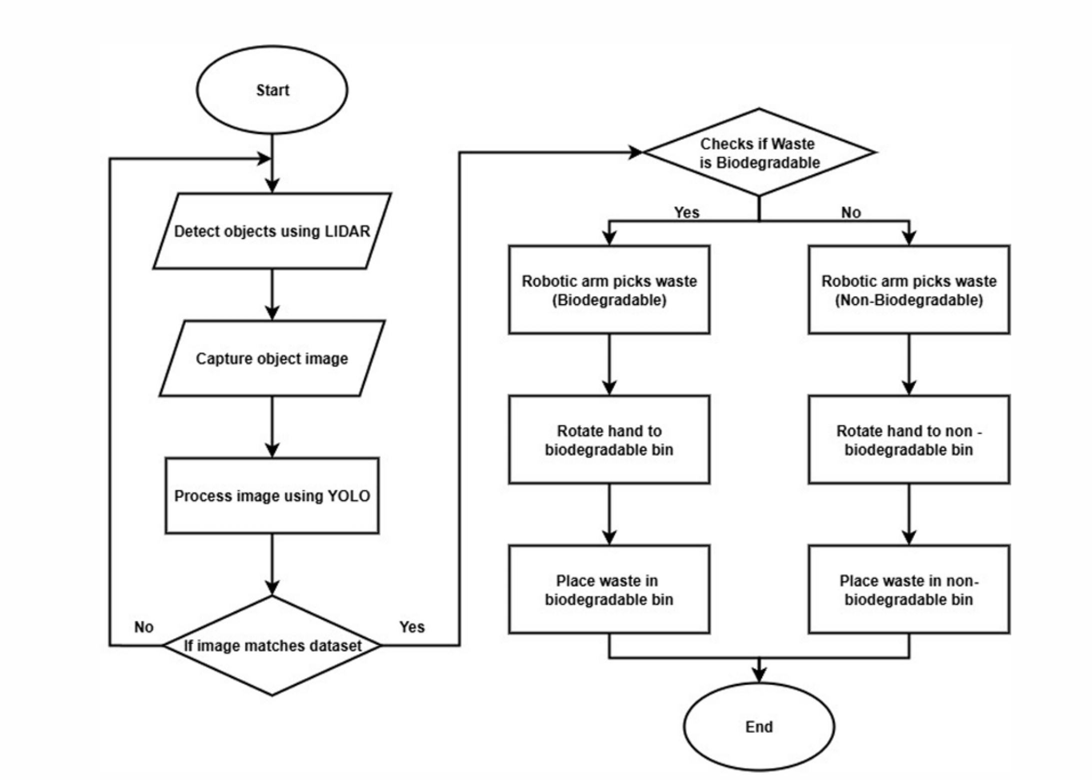


Figure 5.2: Operational flowchart of TrashBot

## 5.4 Summary

This chapter outlined the hardware and software components of the TrashBot system. The integration of sensors, motors, and the YOLOv8-based classification system enables autonomous waste collection and segregation. The system utilizes four DC motors for movement, five servo motors for the robotic arm, and nine lithium-ion batteries for power. The detailed specifications ensure reliable performance in various indoor environments, making it suitable for use in smart institutional spaces and public areas.